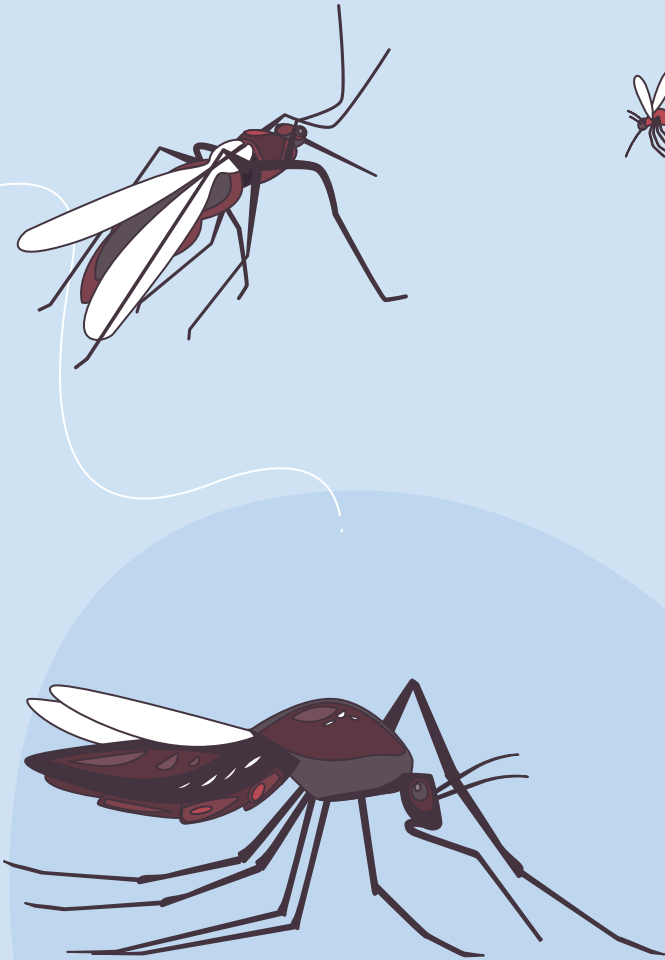




Interesting Case Term infant present with LONS

Pattranan Kusontammarat, MD

Division of Pediatric Infectious Diseases,
Department of Pediatrics
Faculty of Medicine, Chiang Mai University



Case Presentation

Case Term male NB, GA 38 week/LMP, NL, BBW 3570 gm (AGA) APGAR 8-9-10

Date of birth: 25 มิถุนายน 2568 at 10 : 53 pm.



Maternal History

- Age 35 years G 2 P 1001 Last 4 years

- Maternal underlying diseases: no U/D

- Lab ANC: Blood group: B Rh positive

Hb 12/11.5 g/dL Hct 37.7/36 %

Thalassemia screening: not couple at risk for major thalassemia

- Serology: Anti-HIV - Negative

VDRL/ RPR - Negative

- Prenatal US findings: US anomaly at GA 22+6 wk: no gross anomaly



Intrapartum



- **Maternal clinical diagnosed chikungunya Dx 24/6/68**
- Present with high grade fever 1 day PTA มีผื่นแดงขึ้นที่ท้อง ไม่คัน หลังจากนั้นผื่นกระจาย แขนขา เริ่มมีปวดข้อ เข่าสองข้าง ข้อเท้าสองข้าง ข้อศอก ไม่มีไอน้ำมูกเสมหะ ไม่มีคลื่นไส้ อาเจียน ถ่ายเหลว ไม่มีตาแดง บ้วนอยู่อำเภอหาดง แถวบ้านมี chikungunya ระบาด
- Visit ER CMU 24/7/68 work up:
 - CBC Hb 11.8 mg/dl Hct 33.9% WBC 10730 (N 88.9% L 6.1% Eo 0.4% Mn 4.3%) platelet 325,000 cells/cu.mm
 - Hemoculture: NG x II
 - Dengue NS1 Ag negative
- **Impression:** suspected chikungunya infection
- **Management:** supportive treatment paracetamol



Perinatal



Route of delivery: NL. Apgar score at 1 min 8, 5 mins 9, 10 mins 10

PE: Weight 3570 g (P 50-90) Length 52 cm. (P 10-50) Head circumference 33 cm (P90)

Vital signs: BT 37 c. HR 144 /min RR 52 /min BP 77/37 (MAP 50)mmHg SpO2 100

GA: term male infant alert and active .

HEENT: Not pale conjunctiva, no icteric sclera, , AF 1.5 x 1.5 cm, **molding**, PF FT, **caput succedaneum 4x5 cm at parietal area**, no subgaleal hematoma, no cephalhematoma, no CLCP, No preauricular ear pits, **moderate tongue tie**

Heart: Regular, no murmur, normal S1S2 .

Lungs: Clear and equal BSBL, no retraction, no nasal flaring .



Abdomen: No distension, soft, no guarding, no distension, no HSM, no palpable mass, A:V 2:1 .

Neuro: Spontaneous movement, moro/ grasping/sucking reflex positive .

Postnatal

Neonatal jaundice due to suboptimal intake

Management: single phototherapy (26-29/06/68)



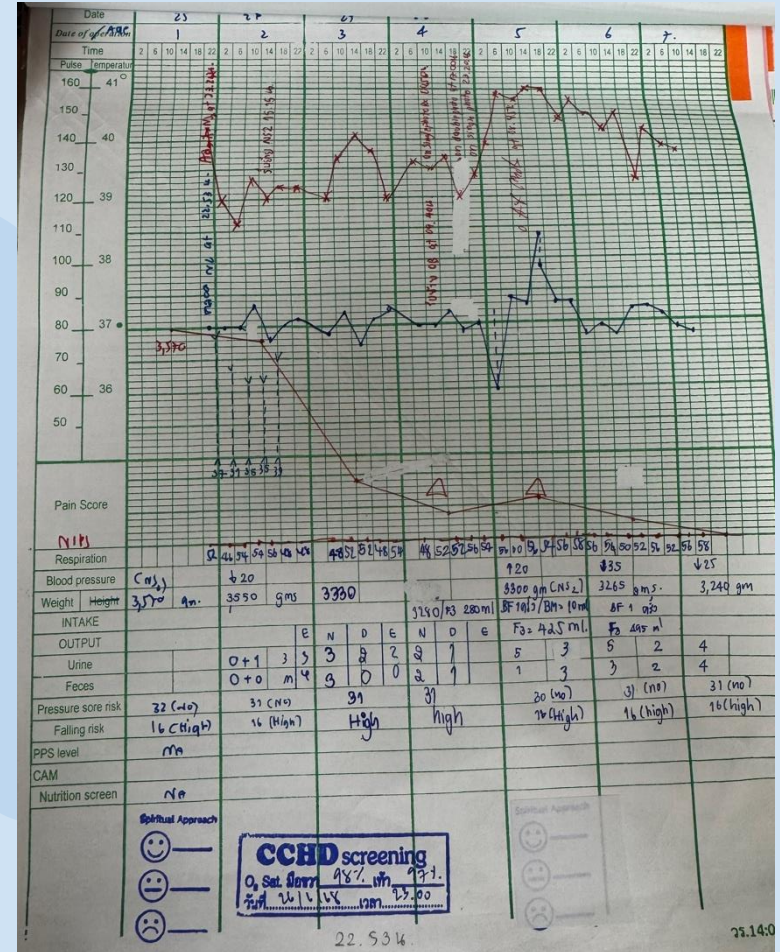
At DOL 5

- Acute fever **BT 38.3 °C**
- No sign of sepsis, no URI
- Septic workup
 - CBC Hb 17.2 mg/dl Hct 48.5% WBC 5,180 (N79.4 L10.6 Baso 0.2%) ALC 550 Platelet 220,000 cells/cu.mm
 - Hs-CRP 1.1 mg/L
 - Nasal swab Flu RSV ATK all negative
 - UA WBC 1-2 HPF, RBC none Ep 1-2 HPF

Impression: AFI suspected viral infection
plan supportive, observe sign of neonatal chikungunya



Mx: request blood for chikungunya PCR



At DOL 7

Develop fever with clinical sepsis

PE: **BT 38.5 °C** HR 140 bpm O2sat 100% RA

Lungs: clear and equal

Ext: CRT 2 secs, **mild palm and sole redness and edema**

Neuro: decreased response, weak cry

Impression: LONS DDx: 1. bacterial sepsis with meningitis

2. Congenital chikungunya with encephalitis

Management: empirical antibiotic ampicillin & cefotaxime meningitis

dose, set lumbar puncture



Lab Work UP



CBC

CBC ...	
☒ Hb	13.5
☒ Hct	38.3
☒ WBC	14500
☒ NRBC per 100 WBC	2
☒ Corrected White Blood Cell	
Differential Per 100 WBC ...	
☒ Blast	
☒ Promyelocyte	
☒ Myelocyte	
☒ Metamyelocyte	
☒ Unidentified cell	
☒ Band	3
☒ Neutrophil	66
☒ Eosinophil	
☒ Basophil	
☒ Lymphocyte	26
☒ Monocyte	3
☒ Atypical Lymphocyte	2
☒ Platelet count	147000
☒ Platelet comment	Adequate
☒ Platelet in smear	Giant platelets were found
☒ RBC	4.29
☒ MCV	89.3
☒ MCH	31.5
☒ MCHC	35.2
☒ RDW	15.6

CSF

WBC	44 cells/cu.mm Corrected WBC 4 cells/cu.mm (PMN 39%, MN 61%)
RBC	2,000 cells/cu.mm
Glucose	57 (ratio 0.7)
Protein	147 mg/dl
Gram stain	no bacteria
Culture	NG

CRP 4.16 mg/L

Blood PCR for Chikungunya



**Office of Disease Prevention
and Control 1 Chiang Mai**
143 Si Donchai Rd. Chang Khlan sub-district, Muang, Chiangmai,
THAILAND 50100, Tel: 053-276 364-108 : Fax: 053-273590



LABORATORY REPORT
ODPC1 No: 6807020003
LSN: 68002955
Accreditation No. 1287/65

[Redacted Patient Information]

Specimen: Serum **Code:** **Objective:** Diagnosis **Event No:**

HIP:** **Request By:** **Collection DT*:** 01-07-2025 ::

Remark: **Received DT*:** 02-07-2025 09:56:24

*Date/Time, ** History of present illness

**** FINAL REPORT ****

Tests	Results	Flag	Unit	Reference Range	Comment
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#Dengue Zika Chikungunya by Real-Time PCR

Dengue	Not detected				
Chikungunya	Detected				
Zika	Not detected				
Detection Method (ZDC)	Real-Time RT PCR by STANDARD TM M10 Arbovirus Panel				



CSF PCR for Chikungunya



Office of Disease Prevention and Control 1 Chiang Mai

143 Si Donchai Rd. Chang Khlan sub-district, Muang, Chiangmai,
THAILAND 50100, Tel: 053-276 364-108 : Fax: 053-273590



LABORATORY REPORT

ODPC1 No: 6807070025

LSN: 68003053



Specimen: Cerebrospinal fluid (CSF)

Code:

Objective: Diagnosis

Event No:

HIP:**

Request By:

Collection DT*: 02-07-2025 ::

Remark:

Received DT*: 07-07-2025 15:26:15

*Date/Time, ** History of present illness

**** FINAL REPORT ****

Tests	Results	Flag	Unit	Reference Range	Comment
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#Dengue Zika Chikungunya by Real-Time PCR

Dengue	Not detected
Chikungunya	Detected
Zika	Not detected
Detection Method (ZDC)	Real-Time RT PCR by Vitassay qPCR Zika Dengue Chikungunya



Impression

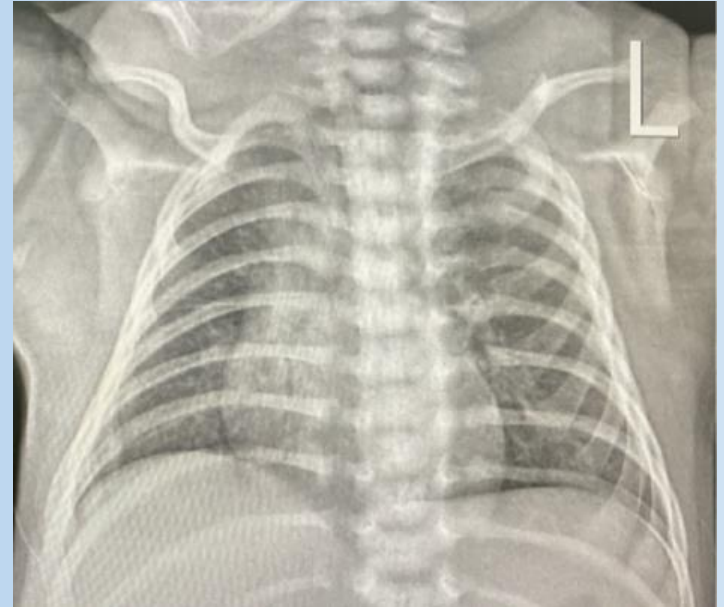
Term male NB, GA 38 week/LMP

Congenital chikungunya with encephalitis



Progression

- At DOL 10 persistent low grade fever & developed clinical dyspnea, need high flow nasal cannula for oxygen support
- **Impression:** HAP
- **Management:** cefazidime, amikacin for total 10 days



Progression

- Developed hyperpigmentation at DOL 15 at perioral, nose and extremities



Progression



Neonatal Chikungunya Infection



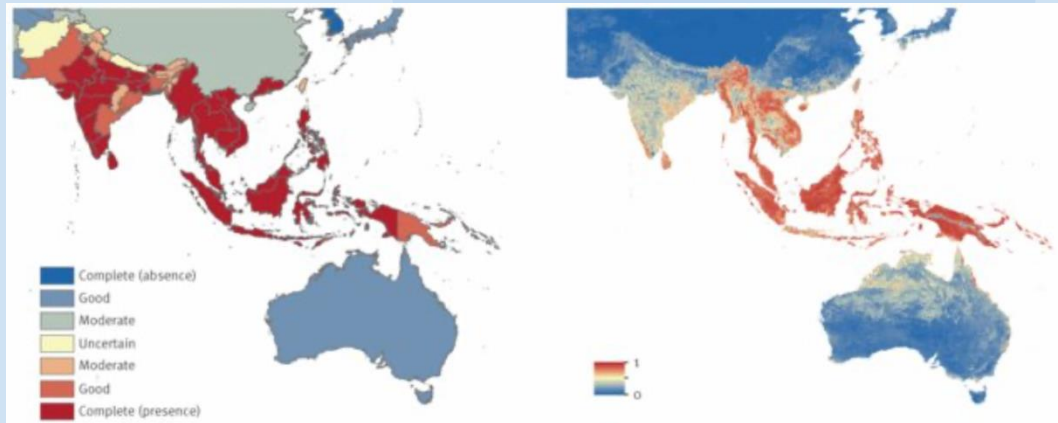
Chikungunya Virus

- RNA virus, Genus Alphavirus, Family Togaviridae
- Transmitted by the *Aedes spp.* mosquitos *A. aegyptii* and *A. albopictus*
- One serotype
- 4 main lineages: West African, **Asian/East/Central/South African (ECSA)** and Indian Ocean Lineage (IOL)
- **Incubation period: 3-7 days (range 1-12 days)**



Epidemiology

- First identified in Tanzania (East Africa) in 1953
- Since then, it has caused outbreaks in Africa, Indian Ocean islands, South East Asia, Central and South America, US territories and Europe
- CHIKV has now been identified in 94 countries worldwide



Transmission

Mode of transmission

1. Mosquito bite: *Aedes aegypti*, *Aedes albopictus*

2. **Vertical transmission**

- intrauterine (rare)
- **intrapartum/peripartum**
- Chikungunya viral RNA has been documented in human milk during an acute chikungunya virus infection, but no infants have been found to be infected through breastfeeding

3. Blood transfusion (rare)



Neonatal Chikungunya

- Vertical transmission of CHIKV was first reported during a large epidemic on Reunion Island in June 2005
- Although fetal infection appears to be extremely rare the **infection rate of newborns from viremic mothers can reach 50%**
- **Up to half of cases experience severe disease and encephalopathy/encephalitis possibly followed by neurological sequelae**
- In the neonate, the most common clinical manifestations are the triad of fever, irritability, and prostration
- Other symptoms in order of frequency are peripheral edema, petechial, and roseoliform rash
- Thrombocytopenia, lymphopenia, and hemolysis have also been observed
- Hyperpigmentation and fine peeling at the extremities were described in newborns at the end of rash



Risk of Mother to Child Transmission

RESEARCH ARTICLE

Mother-to-child transmission of Chikungunya virus: A systematic review and meta-analysis

Despina Contopoulos-Ioannidis^{1*}, Shoshana Newman-Lindsay², Camille Chow³, A. Desiree LaBeaud¹

¹ Department of Pediatrics, Division of Infectious Diseases, Stanford University School of Medicine, Stanford, CA, United States of America, ² Department of Pediatrics, Children's Hospital of Richmond, Virginia Commonwealth University, Richmond, VA, United States of America, ³ Department of Internal Medicine, St. Agnes Medical Center, Fresno, CA, United States of America

- Across 8 cohorts (13 studies, 1,331 pregnancies): overall risk **15.5%** transmission to the fetus/newborn
- Intrapartum exposure** (infection in mothers within ~2 days before or after delivery): **50%** (23/46 infants) developed symptoms **VS** 0% (0/712) among antepartum/peripartum maternal infections

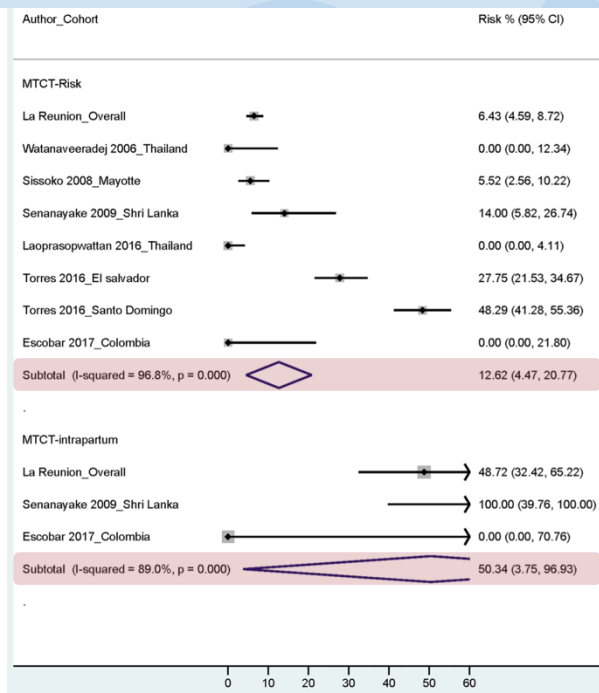
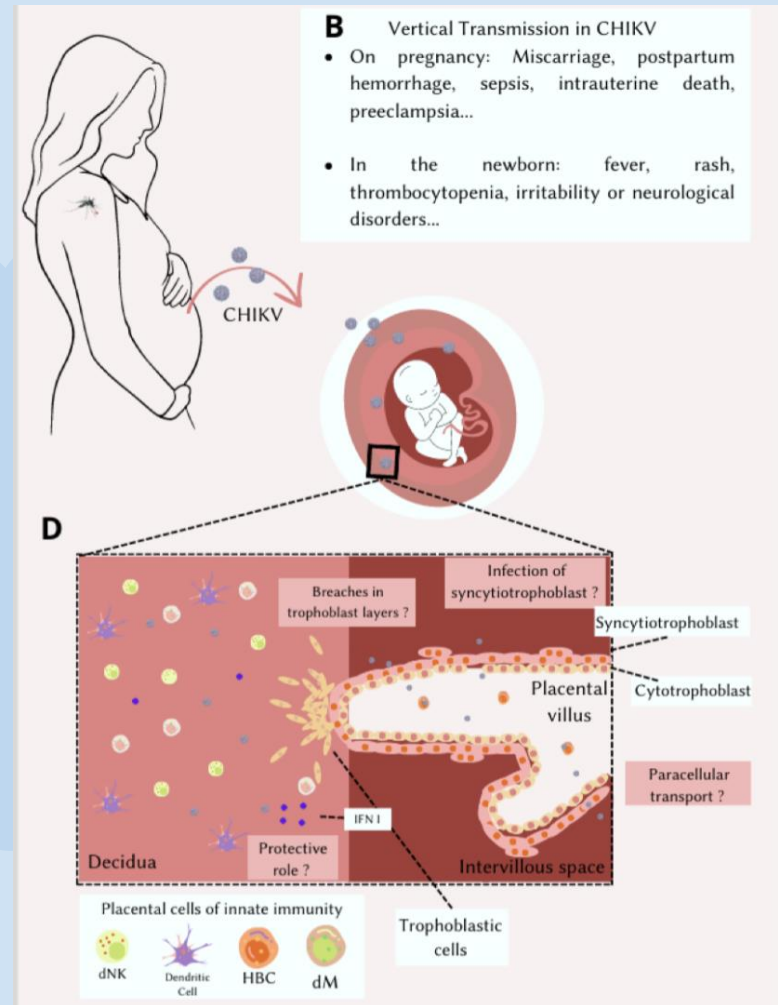


Fig 2. Mother-to-child-transmission (MTCT) risk: Overall and among intrapartum maternal infections. (For studies that did not report data on all (symptomatic and asymptomatic) neonatal infections, we extrapolated the data from the symptomatic neonatal disease cases [27, 51, 55]; and we considered that the MTCT-risk was at least the risk of symptomatic neonatal disease. Most cohorts that reported neonatal infections had reported only symptomatic cases [27, 51, 55] [S3 and S4 Tables]).

Vertical Transmission

- The role of the placenta in CHIKV transmission is not fully understood
- It is believed that vertical transmission can occur via the **direct infection of trophoblasts or syncytiotrophoblasts**, as well as from breaches of the trophoblast layer or via **paracellular transport from maternal blood to fetal capillaries**
- Vertical transmission **occurred regardless of the mode of delivery**, almost always occurring during **maternal viremia in the peripartum period**





Perinatal and Neonatal Chikungunya Virus Transmission: A Case Series

Fátima C. P. A. Di Maio Ferreira¹, Ana M. Bispo de Filippis², Maria Elisabeth L. Moreira³, Simone B. de Campos⁴, Trevon Fuller⁵, Fernanda C. R. Lopes⁶, and Patrícia Brasil⁷

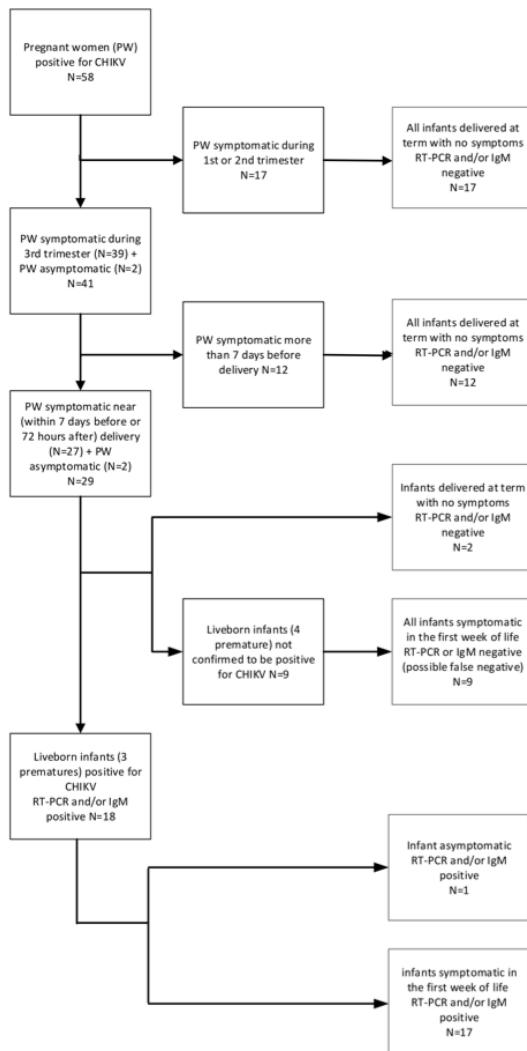
- Newborns with confirmed exposure to CHIKV in utero or in the neonatal period between March 2016 and August 2020 Brazil
- The diagnosis of neonatal CHIKV infection was based on RT-PCR detection of the viral genome in the neonate's serum, CSF, and urine and/or anti-CHIKV IgM in serum and CSF during the first 28 days of life

Table 1. Demographics of Pregnant Participants Infected with CHIKV at a University Hospital During an Epidemic in Rio de Janeiro, Brazil, 2016–2020 (n = 58)

Maternal Demographics (N = 58)	
Maternal age, median (range)	23.2 (16–41)
Maternal Race	
Black or mixed race	43 (74%)
White	15 (25.5%)
Highest Maternal Educational Level	
Primary school	18 (30.5%)
Secondary school	29 (51%)
Higher education	11 (18.5%)
Timing of Onset of Maternal CHIKV Symptoms	
First trimester	9 (15.5%)
Second trimester	8 (14%)
Third trimester	39 (67%)
Clinical Signs and Symptoms	
Any symptoms	57 (96.5%)
Arthralgia (non-inflammatory joint pain)	48 (81%)
Rash	46 (78%)
Fever	45 (76%)
Arthritis (inflammatory joint pain)	16 (27%)
Myalgia	8 (13.5%)
Pruritus	4 (7%)
Cholestasis	3 (5%)
Asymptomatic	2 (3.5%)
Comorbidities	
Any comorbidity	18 (30.5%)
Preeclampsia	6 (10%)
Hypertension	3 (5%)
Urinary tract infection	4 (7%)
Diabetes	2 (3%)
Hashimoto's disease	2 (3%)
Epilepsy	1 (1.5%)
Placenta previa	1 (1.5%)
Rheumatic arthritis	1 (1.5%)
Urinary sepsis	1 (1.5%)

CHIKV—chikungunya virus.

Case Series



- Of the 58 infected pregnant people, 39 experienced symptoms during the third trimester, **with 27 experiencing symptoms within a week before or 3 days after delivery**
- **In the subgroup of pregnant people who were viremic near delivery, the vertical transmission rate was 62% (18/29)**
- Two cases of neonatal CHIKV infection were characterized as postnatal and confirmed as mosquito-borne transmission
- Only one case of neonatal CHIKV infection by vertical transmission was asymptomatic
- The others were severe cases that required admission to the Neonatal Intensive Care Unit (NICU)

Clinical Manifestations

Table 3. Clinical Manifestations Among CHIKV-Infected Newborns ($n = 20$) at a University Hospital During Epidemics in the Period of 2016–2020 in Rio de Janeiro

	Clinical Manifestations	N (%)
Neurological	Hypoactivity Irritability Weak suction Seizure Encephalitis	14 (70)
Fever		11 (55)
Dermatological	Rash Hyperpigmentation Vesicobullous lesions	10 (50)
Respiratory	Apnea Respiratory failure	9 (45)
Hematological	Thrombocytopenia Disseminated intravascular coagulation	8 (40)
Muscular	Hyperalgesia	5 (25)
Cardiovascular	Hemodynamic instability	1 (5)
Total number of children with CHIKV infection		20 (33)

CHIKV—Chikungunya virus.

Table 4. Clinical and MRI Follow-Up Characteristics of Three Newborns with CHIKV Encephalitis at a University Hospital During Epidemics in the Period of 2016–2020 in Rio de Janeiro

	Case 1	Case 2	Case 3
Neonate sex	Female	Female	Female
Gestational age at the onset of maternal symptoms	36 weeks	37 weeks	38 weeks
Clinical and laboratory diagnosis of maternal CHIKV	Fever and rash Positive IgM	Fever and arthralgia Positive RT-PCR	Fever and rash Positive IgM
Delivery	Caesarean	Caesarean	Vaginal
Gestational age at birth	36 weeks	38 weeks	38 weeks
Onset of neonatal symptoms	4 days	5 days	6 days
Neonatal age at first brain MRI	17 days	30 days	25 days
First MRI findings	Subcortical cavitations in the frontal and parietal lobes and areas of restricted diffusion throughout the corpus callosum	Subcortical cavitations in the frontal and parietal lobes with perivascular distribution, without areas of restricted diffusion with thinning of the corpus callosum	Discreet focus of hemorrhage in the right parietal lobe with normal myelination for age
Follow-up brain MRI (interval between MRI)	Normal corpus callosum, regression of subcortical cavitations (2.5 months)	Regression of subcortical cavitations (1 year and 4 months)	Persistence of cerebral hemorrhage focus with normal evolution of myelination (10 months)

Neuroradiol J. 2020 Dec; 33(6):532–7 [14].

CHIKV—Chikungunya virus; MRI—Magnetic resonance imaging; RT-PCR—Real-time polymerase chain reaction.

- Cerebral MRI was performed for 65% (n=13) of symptomatic newborns and was altered in 38.4% (n=5) of the cases
- In 3 children in F/U, a reduction was observed in volume of cavitations, and restricted diffusion were resolved

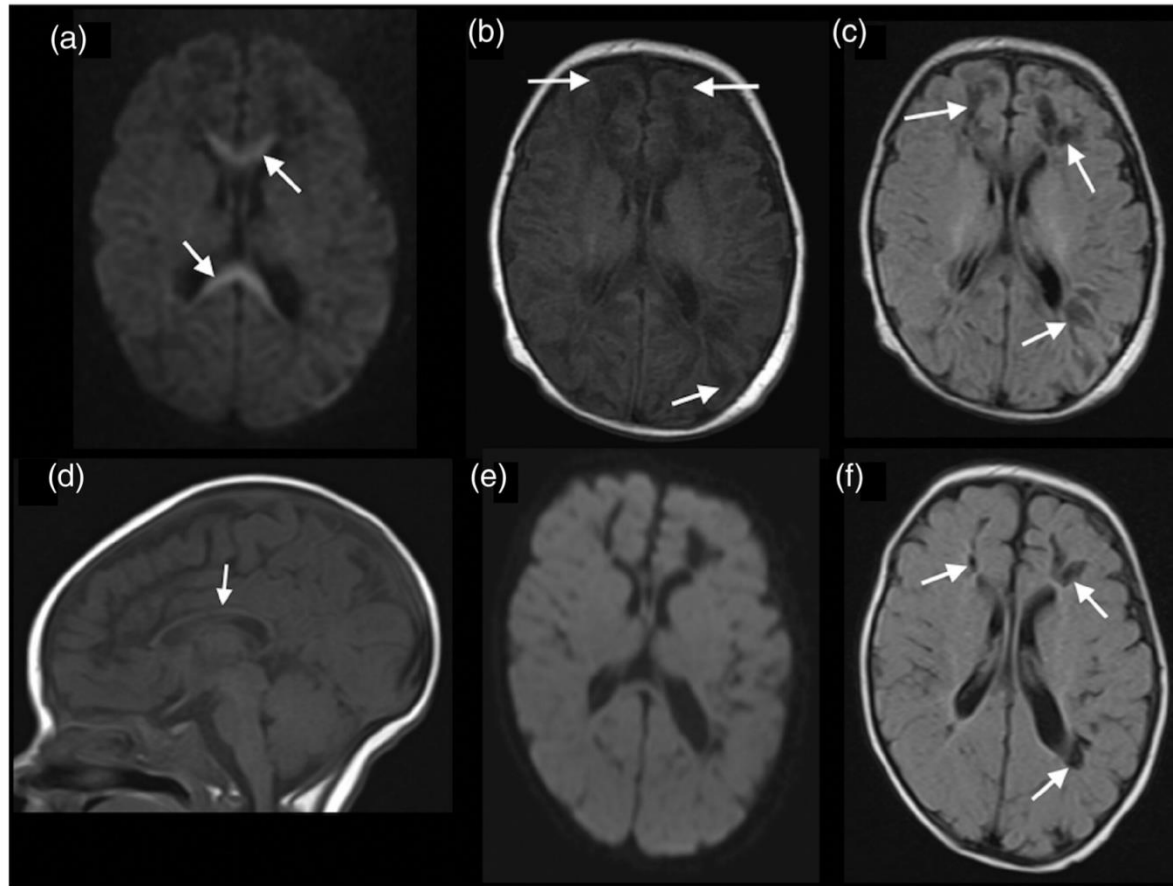


Figure 4. Brain magnetic resonance imaging (MRI) showing restricted diffusion in the corpus callosum (arrows in (A)), associated with subcortical vasogenic edema (arrows in (B)), and cavitations in the frontal and parietal lobes, suggesting a perivascular distribution (arrows in (C)). Follow-up brain MRI, in the third month of life demonstrated thinning of the corpus callosum (arrow in (D)), without restricted diffusion (E) and that the cavitation had shrunk and the subcortical lesions had improved (arrows in (F)) [1, 13].

Mother to Child Chikungunya: Mae Sot Hospital, Thailand 2019



Table 2 Signs and symptoms of newborns suspected of contracting chikungunya virus during pregnancy (N=6)

Signs/Symptoms	Number (percent)
Fever	6 (100%)
Maculopapular rash	6 (100%)
Diffuse/limbs edema	5 (83.3%)
Irritability	5 (83.3%)
Hyperpigmentation	5 (83.3%)
Respiratory distress	5 (83.3%)
Poor feeding	5 (50.0%)
Seizure	2 (33.3%)
Hemodynamic instability	1 (16.7%)

Of the 6 newborns evaluated, all developed fever with concurrent maculopapular rash, around day 4-5 of life

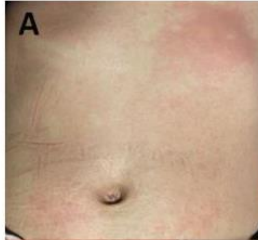


Figure 1, 2 Acrofacial hyperpigmentation on the face, around the mouth and nose (brownie-nose hyperpigmentation) and torso

Cutaneous Manifestation in CHIKV infection



Maculopapular rash



Chik sign: nose hyperpigment



post-inflammatory hyperpigmentation
increased intraepidermal melanin dispersion/retention



Vesiculobullous



Convalescence



Millian's ear sign



Urticaria



Erythema of nose

Erythema of the hand and feet



B

Bullous rash 40% of infants age <6 mo



Figure 3. Clinical manifestations of CHIKV infection in neonates. (A) Post chikungunya pigmentary disorder, (B) chik sign, (C) scaled skin syndrome like presentation, (D) irritability and crusty perioral lesions, (E) roseoliform rash and respiratory distress, and (F) vesicobullous lesions.

Neurological Presentation



Neurologic

ataxia
blindness
brain hemorrhages
CSF hypoglycorachea
CSF protein elevation
EEG abnormalities
encephalitis/ meningoencephalitis
encephalopathy (also with normal CSF chemistry and cytology)
febrile seizures
hydrocephalus and brain infarct
hypotonia
meningeal syndrome
optic atrophy
seizures
status epilepticus
strabismus
Visual Evoked Potentials (VEPs) abnormalities

Long Term Outcome

Neurologic-long term outcome

Normal

global neurodevelopmental delays

behavioral problems

cerebral palsy

conceptual thinking inadequate

hypertonia, spastic diplegia

language skills inadequate

microcephaly

neurodevelopmental disabilities

non-meaningful memory inadequate

nonverbal and numeric reasoning inadequate

numeric reasoning poor

visual-motor skills poor



Management

- No specific antiviral treatment
- Management is supportive with close monitoring of neonates born to infected mothers, especially during the first week of life
- Neonates born to mothers with intrapartum CHIKV infection should be observed for at least 7 days postpartum due to risk of delayed symptom onset
- NSAIDs are contraindicated in pregnancy after 24 weeks due to fetal risks and should be avoided in neonates
- Breastfeeding is generally considered safe as CHIKV transmission via breast milk is not documented



BRIEF REPORT

Long-Term Neurodevelopmental Outcomes of Neonatal Chikungunya: Follow-Up of a Series of Cases Till 1 Year

Amlin Shukla, MD, Tapas Bandyopadhyay, MD, DM,
Nagaratna Vallamkonda, MBBS, PhD and
Arti Maria, MD, DM

- **Primary outcome:** Development Quotient by Developmental Assessment Scales for Indian Infants(DASII)
- Secondary outcome: growth/anthropometric assessment, neuromotor/neurosensory and re-hospitalization rate

Prognosis



- The study was conducted in the high-risk clinic of Neonatology Department of a tertiary referral unit in North India from November 2016 to January 2018
- All of these enrolled neonates at the time of NICU admission presented with features of encephalopathy

Results:

- An important outcome was presence of **neurodevelopmental delay at 12 months of corrected age in over half of babies** including isolated expressive speech disorder in two babies, strabismus in two babies and cortical visual impairment in one baby

Table 1. Baseline characteristics, primary and secondary outcomes of babies with neonatal Chikungunya

Baseline characteristics during hospital stay						Primary outcome	Secondary outcomes at 12 months			
Case No.	Gestational age at birth (weeks)	Birth weight (g)	Gender	Complications during hospital stay	Duration of stay (days)	DASII (12 ± 3 months)	Milestones	Tone	Seizures	Visual assessment
1	36 ⁺⁶	2560	Male	Nil	14	Me-92.0 Mo-102	Age appropriate	Normal	No	Normal
2	37 ⁺⁶	2550	Male	Shock and thrombocytopenia	11	Me-92.4 Mo-102.3	Age appropriate	Normal	No	Normal
3	37 ⁺³	4050	Male	Thrombocytopenia, MODS with shock and respiratory failure	21	Me-97.3 Mo-111	Age appropriate	Normal	No	Normal
4	37 ⁺⁴	2500	Male	Shock and thrombocytopenia	20	Me-99.4 Mo-108.2	Age appropriate	Normal	Yes	Normal
5	35 ⁺⁶	2515	Male	Thrombocytopenia	25	Me-13.3 Mo-13.3	Delayed	Spastic Quadripareisis	Yes	Cortical visual impairment
6	36 ⁺¹	2340	Male	Thrombocytopenia	12	Me-61.6 Mo-43.3	Delayed	Spastic Quadripareisis	No	Strabismus
7	37 ⁺²	2510	Male	Thrombocytopenia	15	Me-76.9 Mo-77.6	Delayed	Hypotonia	No	Strabismus
8	37 ⁺¹	2885	Female	Thrombocytopenia	9	Me-93.8 Mo-102.3	Age appropriate	Normal	No	Normal
9	35 ⁺⁴	2650	Male	Thrombocytopenia	22	Me-99.4 Mo-86.5	Expressive speech delay	Normal	No	Normal
10	38 ⁺⁰	2880	Male	Hypocalcemia and thrombocytopenia	18	Me-86.4 Mo- 87.05	Age appropriate	Normal	No	Normal
11	37 ⁺⁵	2520	Male	Shock and thrombocytopenia	15	Me- 87.7 Mo-90.2	Age appropriate	Normal	No	Normal
12	37 ⁺¹	2535	Female	Thrombocytopenia	8	Me- 87 Mo-93	Age appropriate	Normal	No	Normal
13	36 ⁺⁵	2400	Female	Thrombocytopenia	7	Me-87.05 Mo-86.4	Motor and expressive Speech delay	Normal	No	Normal

Notes: MODS, multiorgan dysfunction syndrome; DASII, Developmental Assessment Scales for Indian Infants; Me, mental score; Mo, motor score.

- DASII was < 85 in three cases
- Out of these three, **two had moderate and one had profound neurodevelopmental delay**
- Two infants who had DQ scores in normal range had isolated expressive language delay

Table 2. Clinico-radiological correlates of abnormalities in imaging done between 15 and 90 days of life (n = 10; 8/10 abnormal)

	Finding on MRI	Age at MRI	Clinical outcome at 12 months of corrected age
1	Diffusion-weighted image showing hyperintense signal in corpus callosum; FLAIR image showing hyperintensity in bilateral centrum semiovale. Suggestive of demyelination due to encephalitis (post-viral).	15 days	Profound neurodevelopmental delay spastic quadriparesis, refractory seizures, microcephaly (West syndrome).
2	CECT—diffuse white matter hyperintensity MRI—normal study.	15 days 3 months	Normal neurodevelopment.
3	Bilateral hyperintensity involving the periventricular and deep white matter of bilateral fronto-parietal region. Cystic changes involving bilateral fronto-parietal and occipital region and dilated lateral ventricles.	20 days 3 months	Moderate delay in mental clusters and severe delay in motor clusters, spastic quadriparesis, convergent strabismus and refractory error.
4	Cystic changes in parieto-occipital periventricular region.	25 days	Moderate delay in motor and mental clusters and convergent strabismus.
5	T2 FLAIR discreet foci of hyperintensities seen in bilateral centrum semi ovale, peritrigonal area and fronto-parietal region suggestive of acute demyelination.	20 days	Normal neurodevelopment.
6	Diffusion restriction in bilateral medial temporal lobe and splenium of corpus callosum likely viral encephalitis.	26 days	Expressive language delay.
7	White matter hyperintensities on T2 and FLAIR images involving frontal and parietal lobes in bilateral periventricular and subcortical region with evidence of diffusion restriction in rostrum and splenium of corpus callosum. Cystic encephalomalacia and ventricular dilatation.	17 days 3 months	Expressive language delay.
8	T2 and FLAIR of frontal and parietal lobe: deep white matter hyperintensities in bilateral peri-ventricular and subcortical region with evidence of diffusion restriction in splenium of corpus callosum. Diffuse cerebral atrophy.	23 days 3 months	Moderate delay in motor and mental clusters, microcephaly, seizures and spastic quadriplegia.

Notes: MRI, magnetic resonance imaging; FLAIR, fluid-attenuated inversion recovery; CECT, contrast-enhanced computed tomography.

Table 3. Studies with data on neurodevelopmental outcomes among babies with congenital Chikungunya infection [15, 16, 18, 19]

S. No.	Study authors	Study type	No. of infants followed up	Study duration	Outcome
1.	Gerardin <i>et al.</i> [15]	Cohort	33 cases of congenital Chikungunya compared with 135 uninfected cases	Average 21 months	Neurodevelopmental delay higher in infected children as compared with uninfected and more in encephalopathic, five developed microcephaly and four labeled as cerebral palsy.
2.	Gerardin <i>et al.</i> [16]	Case series	9 with encephalopathy during initial admission	16–24 months	Four had persistent disabilities: 1 cerebral palsy with ataxia and blindness following extensive white matter degeneration, three had ocular and behavioral or postural deficiencies and one multiple seizures.
3.	Shenoy <i>et al.</i> [18]	Case series	2 congenital Chikungunya cases	36 months	Case 1: spastic diplegia, seizure disorder, borderline IQ (Intelligence quotient) and Case 2: primary optic atrophy, Poor IQ
4.	Ramos <i>et al.</i> [19]	Case report	1 case of congenital Chikungunya	1 year	Microcephaly, delayed development, optic atrophy, seizures.

Thank You

